



PH 603: Advanced Condensed Matter Physics

Course Code	PH 603
Course Name	Advanced Condensed Matter Physics
L-T-P-C	3-0-0-3
Prerequisites	PH301 Quantum Mechanics-I, PH 501 Introductory Solid State Physics or faculty consent.
Intended for	PG and UG
Distribution	Elective
Approval	8th Senate (Oct 2015)

Course Contents

- **Modules I**
- Second quantization for Fermions and Bosons, Review of Bloch's theorem, tight binding Model, Wannier orbitals, density of states. [6 Lectures]
- **Modules II**
- Born-Oppenheimer approximation, Effects of electron-electron interactions - Hartree- Fock approximation, exchange and correlation effects. Fermi liquid theory, elementary excitations, quasiparticles. [6 Lectures]
- **Modules III** [7 Lectures]
- Dielectric function of electron systems, screening, random phase approximation, plasma oscillations, optical properties of metals and insulators, excitons, polarons, fluctuation- dissipation theorem.
- **Modules IV** [7 Lectures]
- Review of harmonic theory of lattice vibrations, anharmonic effects, electron-phonon interaction -mass renormalization, effective interaction between electrons, polarons.
- **Modules V** [8 Lectures]
- Metal-Insulator transition, Mott insulators, Hubbard model, spin and charge density waves, electrons in a magnetic field, Landau levels, integer quantum Hall effect.
- **Superconductivity** [8 Lectures]
- Phenomenology, Cooper instability, BCS theory, Ginzburg-Landau theory.

Text Books

1. N. W. Ashcroft and N. D. Mermin, **Solid State Physics**, Holt, Rinehart and Winston, 1976)
2. C. Kittel, **Quantum Theory of Solids**, Wiley, 1987.
3. M. P. Marder, **Condensed Matter Physics**, John Wiley & Sons, 2010.
4. H. Ibach and H. Luth, **Solid State Physics**, Springer Science & Business Media, 2009.

References

1. W. Jones and N. H. March, **Theoretical Solid State Physics**, Courier Corporation, 1985).
2. Phillips, **Advanced Solid State Physics**, Cambridge University Press, 2012.
3. G. D. Mahan, **Many Particle Physics**, Springer Science & Business Media, 2000.
4. D. Pines, **Elementary Excitations in Solids**, Advanced Book Program, Perseus Books, 1999.
5. Patrik Fazekas, **Lecture Notes on Electron Correlation and Magnetism**, World Scientific, 1999.
6. Giuliani and Vignale, **Quantum Theory of the Electron Liquid**, Cambridge University Press, 2005.